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Spherical Caps and Feature Cones

A Geometric Analysis of Adversarial False-Accepts, SP-030

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Context & Motivation

The adversarial threat in Deep Biometrics

The Success of Deep Biometrics:

Deep Neural Networks (DNNs) have achieved superhuman performance in Face Recognition, leading to widespread deployment (e.g., smartphones, border control).

The Vulnerability:

However, these systems remain highly vulnerable to **Adversarial Attacks**. An attacker can add visually imperceptible noise to their own face to trigger a **False Accept** (successfully impersonating a target identity).

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Context & Motivation

The adversarial threat in Deep Biometrics

The Fundamental Question:

Are adversarial false-accepts simply a result of “flawed training” or “bad data”, or are they a **structural inevitability** dictated by the geometry of high-dimensional spaces?

Our approach: We shift the focus from the pixel space to the topological properties of the latent space (embeddings) to mathematically explain this vulnerability.

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Introduction to biometric verification

Modeling and application of geometric priors

- Input: $x \in \mathbb{R}^{h \times w \times c}$
- Encoder: $f_\theta : \mathbb{R}^{h \times w \times c} \to S^{d-1}$
- Normalized embedding: $z = f_\theta(x) \in S^{d-1}$
- Cosine decision threshold: $\langle z, t \rangle \geq m$

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Introduction to biometric verification

Modeling and application of geometric priors

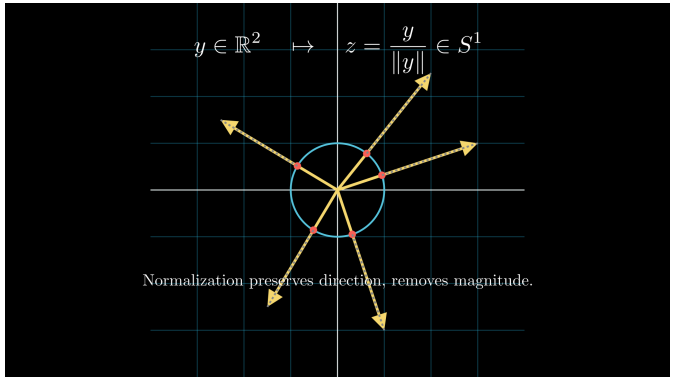

Figure 1: Normalization to S^1.

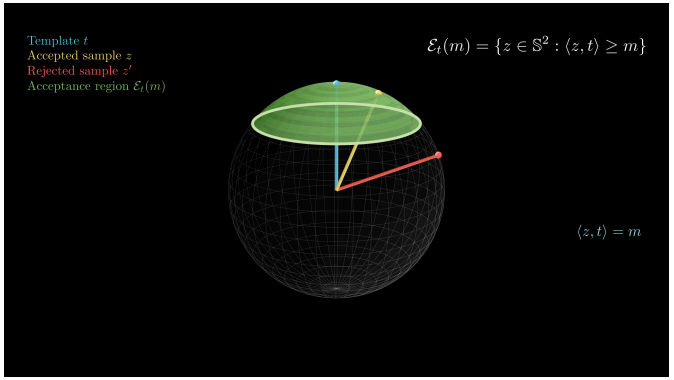

Figure 2: Spherical cap in S^2.

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Introduction to biometric verification

Modeling and application of geometric priors

The acceptance region of a template t is a spherical cap:

$$\mathcal{E}_t(m) = \{z \in S^{d-1} : \langle z, t \rangle \geq m\}$$

Consequence: Adversarial robustness inherently depends on the geometric margin to this cap on S^{d-1}.

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The Isotropic Baseline

The structural pressure of high dimensions

What if impostors are uniformly distributed on the sphere?

Theorem 1 (Gaussian Approximation)

In high dimensions, the inner product $\langle X, u \rangle$ concentrates around 0:

$$\boxed{\mathrm{FAR} \approx 1 - \Phi\left(m\sqrt{d}\right)}$$

Theorem 2 (Average distance to the cap)

The expected Euclidean distance d_{emb} between an impostor and the acceptance zone is:

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The Isotropic Baseline

The structural pressure of high dimensions

$$\mathbb{E}[d_{\text{emb}}] \approx \frac{\alpha\Phi(\alpha) + \varphi(\alpha)}{\sqrt{d}} \quad \text{where } \alpha = \Phi^{-1}(1 - \text{FAR})$$

- **Scaling:** The safety margin collapses in $\mathcal{O}\left(\frac{1}{\sqrt{d}}\right)$.
- **Instability:** Even for extremely low FAR (e.g., 10^{-6}), geometric margins are tiny in high dimensions (e.g., $d = 512$).

Beyond Isotropy

Anisotropy and Feature Cones

Real biometric embeddings are **not** uniform. They cluster in tight regions, leaving large parts of the sphere empty ("Geometric Sparsity").

Feature Cone Model

Identities cluster in cones defined by a centroid μ_k and opening ω_k:

$$\mathcal{C}_k(\omega_k) = \{z \in S^{d-1} : \langle z, \mu_k \rangle \geq \cos(\omega_k)\}$$

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Beyond Isotropy

Anisotropy and Feature Cones

The Anisotropy Gap (B_t)

This clustering acts as an “Anisotropy Shield”, artificially inflating the distance to errors compared to the theoretical isotropic baseline.

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Adversarial Reachability

From Embedding geometry to Pixel perturbations

How does the embedding-space distance d_{emb} translate to actual image perturbations δx?

The Local Jacobian Bound

Assuming the encoder f_θ is locally $L(x)$-Lipschitz:

$$\left\| f_{\theta(x+\delta x)} - f_{\theta(x)} \right\|_2 \leq L(x) \|\delta x\|_2$$

Therefore, any successful adversarial attack must satisfy:

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Adversarial Reachability

From Embedding geometry to Pixel perturbations

$$\|\delta x\|_2 \geq \frac{d_{\text{emb}}(z, \mathcal{E})}{L(x)}$$

Takeaway: Robustness requires **both** favorable geometry (large d_{emb}) and strict control of the encoder's sensitivity (small $L(x)$).

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Experiments (1/2): Synthetic Validation

Validating the isotropic model on $S^(d-1)$

Protocol:

- Synthetic biometric model on S^{d-1}
- EER fixed at 0.7% across $d \in \{64, 128, 256, 512\}$
- Measure distance-to-error boundaries

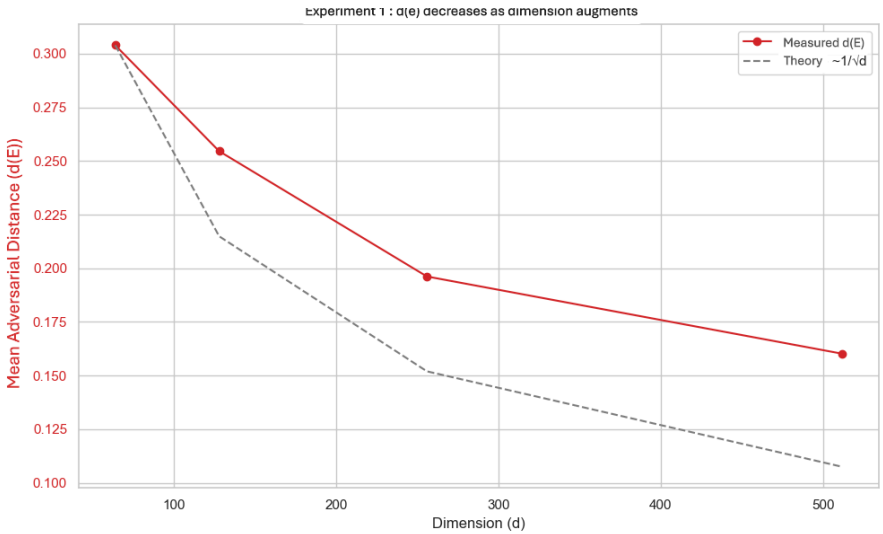

Figure 3: Empirical d_{emb} drops from ≈ 0.30 ($d = 64$) to ≈ 0.16 ($d = 512$).

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Experiments (1/2): Synthetic Validation

Validating the isotropic model on $S^(d-1)$

Result: The empirical distance perfectly follows the $\mathcal{O}\left(\frac{1}{\sqrt{d}}\right)$ scaling predicted by Theorem 2.

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Experiments (2/2): IJB-C Benchmarks

Measuring the Anisotropy Shield

Real Face Embeddings (IJB-C):

- Theory predicts: $d_{\text{emb}} \approx 0.2$
- Real data shows: $d_{\text{emb}} \approx 0.7 - 0.9$

Conclusion:

The "Anisotropy Shield" factor is $B \approx 3.0$. Real systems are artificially protected by data clustering, mitigating the curse of dimensionality.

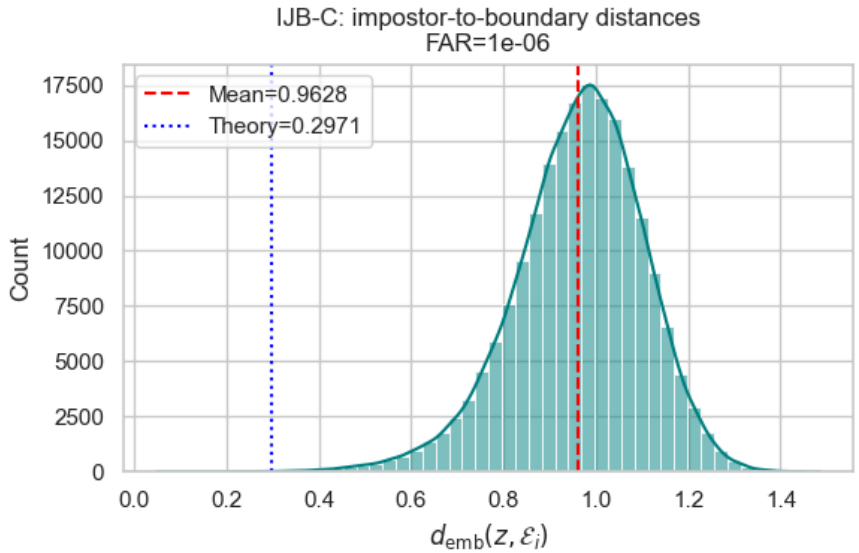

Figure 4: Adversarial margins on IJB-C vs Isotropic Baseline.

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Conclusion & Takeaways

The tension between geometry and empirical data

1. **The Isotropic Baseline:** High dimensions structurally push adversarial false-accept margins toward zero ($\frac{1}{\sqrt{d}}$ scaling).
2. **The Anisotropy Shield:** Real data isn't uniform. "Feature cones" provide a natural safety margin 3x larger than the baseline.
3. **Adversarial Feasibility:** Despite this shield, attacks remain feasible because they exploit the local sensitivity (Jacobian) of the deep neural network.

Future Work: Expanding this geometric framework to composite pipelines (e.g., Face Recognition + Presentation Attack Detection).

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Thank you for your attention

Questions & Discussion

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Paper: Spherical Caps and Feature Cones: A Geometric Analysis of Adversarial False-Accepts

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